Pedro Henrique Rocha Moy | phrmoy@gmail.com

Artificial Intelligence, Machine Learning, Data Science, Data & Software Engineering

github.com/phrmoy • linkedin.com/in/phrmoy/ • rocha-moy-engineering-technology.github.io/resume/

Belo Horizonte, Brazil, America/Sao_Paulo, BRT (UTC-3) • +55 31 99452-9351 Miami, United States, America/New_York, EST (UTC-5) • +1 508-933-9351

Citizenships: US & Brazilian • **Fluency:** English & Portuguese

Education

Executive MBA in Business Administration

University of Miami - Miami, 2021 - 2022

Master's Degree in Computer Science (Machine Learning)

Georgia Institute of Technology - Atlanta, GA, 2015 - 2017

Master's Degree in Earth Science and Engineering (Geophysics)

King Abdullah University of Science and Technology - Saudi Arabia, 2010 - 2012

Bachelor's Degree in Mechanical Engineering (Math Minor)

University of Massachusetts Lowell - Lowell, MA, 2008 - 2010

Experience

AI Advisor | Toptal Client | 2026

- Partnered with managers and engineers to assess the current state of the company's AI pipeline, surfacing its pitfalls and inefficiencies
- Researched and delivered battle-tested strategies, from quick wins to major long-term initiatives, improving the consistency and accuracy of the existing AI pipeline
- Delivered the final AI pipeline design with robust safeguards against missing critical business paths, cutting the size of the AI's instructions by over 90%
- Technologies: Large Language Models (LLMs), AI Architecture, Prompt Engineering, Systems Optimization, Business Services

Head of AI | Toptal Client | 2026

- Devised the proof of concept (POC) and minimum viable product (MVP) strategy and documentation for a new AI-driven technology
- Designed and oversaw a series of case studies and experiments that culminated in a successful POC
- Drove the POC and MVP work in direct partnership with the CEO, overseeing business requirements, technical feasibility, and economic feasibility

- Operated as both leader and hands-on engineer to generate results, directing a PhD-level engineer on technical execution while aligning business strategy with the CEO
- Technologies: Large Language Models (LLMs), Artificial Intelligence (AI), Reinforcement Learning from Human Feedback (RLHF), ChatGPT, Prompt Engineering, Product Strategy, POC/MVP Development, Technical Leadership

AI Consultant | Toptal Client | 2025

- Delivered a prototype platform that automated LLC formation workflows, grounded in a detailed regulatory study for Florida, cutting manual research time for new founders by over 50%
- Built an AI-driven validation step that vetted user inputs and flagged compliance gaps during the LLC creation process, ensuring the accuracy of filings
- Designed a domain-specific knowledge base and an intelligent AI search system that delivered precise, context-aware guidance to users
- Technologies: Python, LLMs, RAG, NLP, Prompt Engineering, Vector Databases, Knowledge Base Design, Document Processing, Regulatory Compliance, Prototype Development

AI Software Developer | Toptal Client | 2023 - 2024

- Built from scratch a platform that converted contract requirement documents into structured suggestions and clarification questions, enabling teams to significantly accelerate project scoping
- Engineered self-scaling cloud infrastructure that sustained stable, high-throughput processing of large procurement documents with near-zero downtime
- Owned the end-to-end AI architecture and served as the primary expert on all AI matters, improving extraction accuracy and capabilities while shaping model-driven workflows across the full product
- Technologies: Python, JavaScript, Django, React, OpenAI API, LLMs, NLP, Prompt Engineering, AWS, Amazon ECS, Infrastructure as Code (IaC), Docker, PDF Processing, Autoscaling, Full-stack Development

Lead Data Scientist | Deloitte Contractor | 2021 - 2022

- Designed and delivered multiple natural language processing models through to production deployment
- Partnered with stakeholders to define use cases and chart the pathway from deployed models to product development and implementation
- Mentored and developed the junior data scientists on the team
- Technologies: Python, NLP, Machine Learning, Model Deployment, Stakeholder Management, Data Analysis, Mentoring

Enterprise Lead Data Architect | Toptal Client | 2020 - 2022

- Owned the architecture, development, and automation of the enterprise's cloud data platform, spanning distributed processing pipelines and data storage
- Delivered automated, scalable cloud infrastructure that responded elastically to development and consumer demand
- Co-managed and supervised a team of engineers end to end, from designing and delegating tasks to mentoring and overseeing delivery

- Technologies: Python, AWS, Amazon EMR, Spark, Distributed Computing, Data Lake Architecture, Cloud Infrastructure, Data Pipelines, Data Storage, Team Leadership

Enterprise Senior ETL and Data Engineer | Toptal Client | 2019 - 2020

- Owned the design and production delivery of the company's large-scale distributed data processing pipelines
- Integrated data across disparate sources and destinations, from raw files and warehouse tables to document collections and real-time streams
- Served as the team's senior engineer and tech lead, strengthening engineering processes, raising software quality control, and shaping sprint planning
- Technologies: Scala, Python, Spark, MongoDB, Hive, Kafka, Distributed Computing, ETL, Data Pipelines, Agile/Scrum, Code Review, Tech Leadership

Hadoop Proof of Concept for Atmospheric Sciences Project | Toptal Client | 2019 - 2020

- Set up the framework for massive distributed computing on the client's existing in-house cluster
- Designed and implemented both general-purpose and specialized data architectures that met the client's query complexity and performance needs
- Delivered accessible software layers that let the client's own teams build on top of the new infrastructure
- Technologies: PySpark, Python, Hadoop, HDFS, Big Data, Data Architecture, Cluster Management, Atmospheric Sciences

Research Data Engineer | Nicklaus Children's Hospital | 2018 - 2019

- Advanced the research department's analytical and data workflows, establishing engineering best practices across its user base
- Delivered both ad hoc and systematic big data pipelines, including validation and integration of varied data sources
- Represented the research department to the IT and business intelligence departments, guiding decisions on analytical and data needs
- Technologies: Spark, Scala, Python, R, Impala, ETL, Big Data, Healthcare Data, Data Validation, Business Intelligence

Technical Advisor | Insight Data Science | 2018

- Advised program fellows on their data engineering projects across problem definition, systems architecture, and execution
- Guided technology selection and architecture for distributed data platforms suited to each project's scale and goals
- Coached fellows through mock interviews covering scalability concepts, algorithms, and computer science fundamentals
- Technologies: Spark, Kafka, Redis, HBase, Cassandra, PostgreSQL, Distributed Systems, Systems Architecture, Mentoring, Algorithms

Senior Software Engineer | NexHealth | 2016 - 2017

- Developed and deployed software at client sites to automate data collection and server synchronization
- Integrated electronic medical records from client systems back to company servers across both database and web channels
- Created a smart text-message response system that let users interact with the company's products entirely via SMS
- Technologies: Python, Scala, JavaScript, Redis, PostgreSQL, Apache Spark, REST APIs, Data Integration, SMS/Messaging, Electronic Health Records, Web Development

Data Scientist | QuaEra Insights | 2016

- Led data science for a consulting engagement, overseeing data management and modeling strategy
- Applied natural language processing to turn unstructured data into actionable business intelligence
- Built a recommendation engine, expressed as business rules, that determined the product was unnecessary in 50% of cases, unlocking substantial cost savings
- Technologies: Python, NLP, Machine Learning, Recommendation Systems, Feature Engineering, Business Intelligence, Data Management

Data Engineering Fellow | Insight Data Science | 2015

- Built themidgame-tube, a platform designed to discover YouTube influencers for brands worldwide
- Deployed a cloud-based big data platform that collected and processed billions of data records
- Proved linear scalability in performance testing across a cluster of up to 20 machines
- Technologies: Python, AWS, Apache Spark, EMR, HBase, Big Data, Distributed Computing, Data Ingestion, Scalability Testing

Data Analyst | Cartesian | 2015

- Supported managed analytics engagements, promoting best practices in batch workflows and data management
- Conducted independent research into big data workflows, spanning data mining and business intelligence integration
- Built lightweight data pipelines that gathered, transformed, and delivered external data into business intelligence tools
- Technologies: Python, PostgreSQL, REST APIs, ETL, Data Pipelines, Business Intelligence, Data Mining, Data Analytics

Data Analytics Engineer | Daktari Diagnostics | 2013 - 2015

- Served as lead developer of the company's primary data processing and data analysis applications for desktop platforms
- Developed a calibration model for the company's CD4 diagnostic device, improving system accuracy by 20-30%
- Shipped machine learning models embedded in standalone end-user applications for automated data classification

- Technologies: Python, Machine Learning, Statistical Modeling, Calibration, Data Classification, Desktop Applications, Medical Devices, Data Processing

Portfolio

Pastoral Conscience AI

https://rocha-moy-engineering-technology.github.io/pastoral_conscience_site/

Built an "Artificial Conscience" AI system that generates scripture-grounded spiritual reflections using RAG over the Psalms and DSPy-based conscience checkers. Designed and implemented a governed reasoning pipeline with three verification layers: Helpfulness, Psalm Grounding, and Citation Integrity. The system enforces strict alignment of beliefs; every response is scored, verified, and corrected before delivery. Architecture: Go backend with hexagonal architecture, Gemini File Search API as the retrieval layer, Python and DSPy services for conscience checkers with model fallback, Next.js frontend with real-time SSE progress updates, and PostgreSQL for data persistence.

Cecilia

https://mariacecilia.app

Built Cecilia, a cycle-awareness app for women trying to conceive or planning their family through fertility-awareness methods, released in open test on iPhone, Android, and the web in English, Spanish, and Brazilian Portuguese. Implemented two forecasting methods: a Calendar method that estimates the fertile window from cycle history, and a Symptothermal method that combines cycle timing with basal body temperature and cervical mucus signs, confirming the post-ovulation phase only after both rules are met. Users log periods, temperature, cervical mucus, ovulation tests, and notes, and see daily fertility status, a calendar of recorded and predicted days, and a temperature chart with coverline. Architecture: Go API with PostgreSQL, WorkOS authentication, a Flutter and Riverpod client for iOS, Android, macOS, and web, Railway Serverless hosting, and a static multilingual Astro website on Cloudflare Pages.